

Embedded systems exp 29

```
#include <systemc.h>

SC_MODULE(priority_interrupt) {

    sc_event low_int, high_int;

    void low_ISR() {
        while(true) {
            wait(low_int);
            cout << "Low Priority Interrupt at "
                << sc_time_stamp() << endl;
            wait(3, SC_SEC);
        }
    }

    void high_ISR() {
        while(true) {
            wait(high_int);
            cout << "High Priority Interrupt at "
                << sc_time_stamp() << endl;
            wait(1, SC_SEC);
        }
    }

    void generator() {
        wait(1, SC_SEC);
        low_int.notify();
        wait(1, SC_SEC);
        high_int.notify();
    }

    SC_CTOR(priority_interrupt) {
        SC_THREAD(low_ISR);
        SC_THREAD(high_ISR);
    }
}
```

```

SC_THREAD(generator);

}

};

int sc_main(int argc, char* argv[]) {

    priority_interrupt obj("Priority_INT");

    sc_start(10, SC_SEC);

    return 0;

}

```

The screenshot displays the EDA Playground web interface. On the left, a sidebar contains a 'Languages & Libraries' section with 'C++/SystemC' selected, and a 'Tools & Simulators' section with 'C++' selected. The main area is divided into two code editors: 'testbench.cpp' on the left and 'design.cpp' on the right. The 'testbench.cpp' editor contains the C++ code shown in the previous block. The 'design.cpp' editor contains placeholder comments. Below the editors is a terminal window showing the compilation and execution output. The output indicates that the code was compiled successfully and is running on a SystemC 2.3.3-Accellera platform. The terminal output includes the following text:

```

[2026-08-30 10:24:05 UTC] g++ -o sim -lsystemc *.cpp && echo "Compile done. Starting run..." && ./sim
Compile done. Starting run...

SystemC 2.3.3-Accellera --- May 23 2020 14:33:56
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Low Priority Interrupt at 1 s
High Priority Interrupt at 2 s
Done

```